

Authentication Mode

[Tenant Authentication \(Recommended\)](#)

[Single-Tenant Authentication \(Outdated, Not Recommended\)](#)

[Cross-Tenant Authentication \(Outdated, Not Recommended\)](#)

Parent Topic: [Interface Reference](#)

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Tenant Authentication (Recommended)

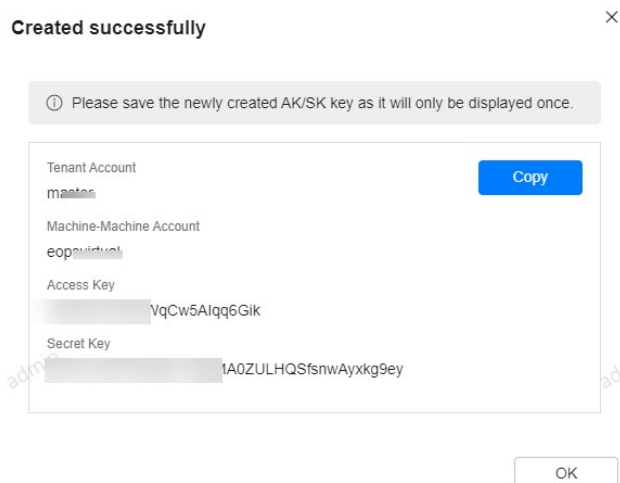
Application Scenarios

When an external program needs to invoke an IMOC interface and obtain the data of a tenant, create a machine-machine user as the tenant administrator (machine-machine users created under the **master** tenant can be used by other tenants) to obtain the required AK/SK. Then, use the AK/SK to obtain the authentication token. With the authentication token, the service interface can now be invoked.

Procedure

1. Log in to IMOC (<https://IP address or domain name for logging in to IMOC/esf-web/>) using an administrator account of a service tenant (for example, the administrator account of the **HWCloudInfra** tenant is **hadmin001**).
2. Click  in the upper right corner of the page, choose **System Management > Integration > Interconnection Account Management**, and select the **Machine Users** tab.
3. Obtain the AK/SK.

Create a machine-machine user on the **Machine-Machine User** tab page to obtain the AK/SK. The AK/SK information is displayed only once. Make a note.



4. Obtain the authentication token.
 - a. Invoke the interface for obtaining a machine-machine token to obtain the authentication token. For details about interface invoking, see [Obtaining a Machine-Machine Token](#).
 - b. When invoking the service interface, transfer the **imoc_token** to the header as the authentication information.

Table 1 Table of header parameters

Name	Value	Description
imoc_token	imoc_token	The imoc_token must be carried when the interface is invoked. For details about how to obtain the imoc_token , see Obtaining a Machine-Machine Token . The validity period is 30 minutes.

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Single-Tenant Authentication (Outdated, Not Recommended)

Application Scenarios

When an external program needs to invoke an IMOC interface and obtain the data of a tenant, the single-tenant authentication mode can be used.

Procedure

1. Log in to IMOC (<https://IP address or domain name for logging in to IMOC/esf-web/>) using an administrator account of a service tenant (for example, the administrator account of the **HWCloudInfra** tenant is **hadmin001**).

- Click  in the upper right corner of the page, choose **System Management > Integration > Interconnection Account Management**, and select the **Machine Users** tab.
- Obtain the AK/SK.

Create a machine-machine user on the **Machine-Machine User** tab page to obtain the AK/SK. The AK/SK information is displayed only once. Make a note.

- iam_user**: The user must have the permission to invoke APIs. The machine-machine user account in the following figure is an example.
- iam_akvalue**: See **Access Key** in the following figure. When invoking the interface, you need to use Base64 to convert the code on any Linux host. The command is as follows:

```
echo '{"user":"machine-machine user account","_tenant":"tenant to which the machine-machine user belongs","gam_ak":"Access K
```

- skvalue**: See **Secret Key** in the following figure.
- iam_tenant**: The tenant to which the machine-machine user belongs, for example, **HWCloudInfra**.

Created successfully

×

① Please save the newly created AK/SK key as it will only be displayed once.

Tenant Account	master	Copy
Machine-Machine Account	eeopwirtual	
Access Key	VqCw5Alqq6Gik	
Secret Key	1A0ZULHQSfsnwAyxkg9ey	

OK

- Obtain the authentication token.
 - Invoke the **iam_token** interface to obtain the authentication token. For details about how to call the interface, see [Obtaining iam_token \(Single-Tenant Authentication\)](#).
 - For single-tenant authentication, transfer the **iam_token** to the header as the authentication information.

Table 1 Table of header parameters

Name	Value	Description
iam_token	iam_token	The iam_token must be carried when the interface is invoked. For details about how to obtain the iam_token , see the external interface of the gam-adapter . The validity period is 30 minutes.

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Cross-Tenant Authentication (Outdated, Not Recommended)

Application Scenarios

A machine-machine user can invoke interfaces of multiple service tenants and obtain data and resources of multiple tenants.

Procedure

- Log in to IMOC (<https://IP address or domain name for logging in to IMOC/esf-web/>) as the **admin** user. Perform the following operations as the **master** tenant.
- Click  in the upper right corner of the page, choose **System Management > Integration > Interconnection Account Management**, and select the **Machine Users** tab.
- Obtain the AK/SK.

Create a machine-machine user on the **Machine-Machine User** tab page to obtain the AK/SK. The AK/SK information is displayed only once. Make a note.

- iam_user**: The user must have the permission to invoke APIs. The machine-machine user account in the following figure is an example.
- iam_akvalue**: See the Access Key in the following figure. When invoking the interface, you need to use Base64 to convert the code on any Linux host. The command is as follows:

```
echo '{"user":"machine-machine user account","_tenant":"tenant to which the machine-machine user belongs","gam_ak":"Access K
```

- skvalue**: Secret Key in the following figure
- iam_tenant**: the tenant to which the machine-machine user belongs, for example, **tenant 1**

Created successfully



① Please save the newly created AK/SK key as it will only be displayed once.

Tenant Account	master	Copy
Machine-Machine Account	ecp	
Access Key	VqCw5Alqq6Gik	
Secret Key	1A0ZULHQSfsnwAyxkg9ey	

OK

- **iam_user**: The user must have the permission to invoke APIs. The machine-machine user account in the following figure is an example.

4. Obtain the authentication token.

- The **iam_token** is obtained using the **AssumeRole** API. For details, see [Obtaining iam_token \(Cross-Tenant Authentication\)](#).
- When invoking each service interface, you must transfer the parameters listed in the following table to the request header.

Table 1 Table of header parameters

Name	Value	Description
iam_token	iam_token	The iam_token is obtained by invoking the AssumeRole interface. The validity period is 30 minutes.

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